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Predicting and analyzing ferry transit delays using open data and machine learning

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ABSTRACT

The utilization of public transport data has evolved rapidly in recent decades. Ferries, with their unique characteristics and sensitivity to weather conditions, pose significant challenges for delay prediction. Given their pivotal role in the transportation systems of numerous cities, accurately predicting ferry delays is crucial for synchronizing transit services.

This paper demonstrates the value of open data for improving ferry delay predictions through machine learning, focusing on two case studies. Our approach leverages General Transit Feed Specification (GTFS) data, ridership and vessel information, and hourly weather data, combined with SHAP explainable artificial intelligence analysis to assess key delay determinants. While support vector regression and deep neural networks showed high accuracy in individual case studies, gradient boosting consistently offered the best balance between prediction accuracy and computational efficiency. Moreover, SHAP analysis reveals that operational and temporal features – such as stop sequence, trip start time, headway, and vehicle label – are the dominant drivers of delays, with weather-related factors exerting only a modest influence.

1. Introduction

Public transportation plays an increasingly important role in modern transportation systems and has emerged as one of the most crucial modes of transportation due to its positive environmental and cost impacts. The importance of public transport relies on providing reliable service and predicting delays so that passengers are informed beforehand.

Public transport modes can be divided into buses, rail (including metros and trams), ferries, and beyond (e.g., cable cars, demand-responsive transit with cars). While most research on delay and travel time predictions for transit focuses on buses and rail services (as evidenced by the fact that all papers included in the survey by Ge et al. (2021) are about these modes), it is important to note that ferry transit is also a significant mode of transportation. For instance, the Sydney ferry network in Australia carries over 15 million passenger journeys annually,[1] the New York (NYC) ferry system saw a total of 6.8 million riders in 2023,[2] and the Hamburg HADAG ferry system in Germany serves approximately ten million passengers annually.[3]

In the ideal case, when focusing on scheduled services, the intended arrival and departure times provided by agencies should match the actual times. Nevertheless, delays often occur due to several external and unforeseen factors, which transit agencies must manage. Analyzing and predicting the delay of a ferry is a challenging task due to the various factors that can affect it. Ferries – whether ships, watercraft, or amphibious vehicles – experience travel time variations caused by many dynamic influences. Despite the challenges in predicting ferry delays, there have been significant advances in this area in recent years. Accurate prediction of ferry delays is essential for the maritime industry, as it enables more precise estimations of expected arrival times, significantly impacting overall planning, as indicated in Cammin et al. (2020).

In particular, the use of machine learning (ML) in both public transport and the maritime industry has gained considerable momentum

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[1] Sydney Ferry network data: www.transport.nsw.gov.au.
[2] NYC ferry system data: https://ny1.com/nyc/all-boroughs/transit/2023/12/15.
[3] Hamburg HADAG ferry system data: https://hadag.de/en/branding/.

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due to the large amount of data generated by these transportation modes. Recent advancements in ML algorithms have made it possible to effectively analyze and make real-time predictions based on this data (Cammin et al., 2020). Indeed, ML algorithms such as support vector regression (SVR), random forests (RF), gradient boosting (GB), and deep neural networks (DNN) can be used to predict vessel delays or arrival times and address the challenges in this area. These models can incorporate various internal and external factors, including agency data, ridership information, and weather conditions. This information enables organizations to make more informed decisions and reduces uncertainty about the delays of ships.

One of the key issues in ferry transit is its synchronization with other transport modes, and accurate predictions can pave the way to improve the transit network schedule (Ricard et al., 2022). Additionally, recent developments in ML, such as the analysis of feature importance, aim to improve model interpretability and understand the impact of different features.

In this paper, we demonstrate that it is possible in the public transport domain to integrate openly available data sources to support the expanded use of ML in public transport planning. We adopt proper feature engineering techniques in addition to recent trends in explainable and interpretable ML. This approach helps to achieve the most accurate prediction of delays and understand the factors that affect them.

In summary, this paper presents three significant contributions that, to the best of our knowledge, have not been applied previously in the context of ferry transit:

- Our proposal introduces a unified approach that combines internal data sources provided by the agency and external data sources (hourly weather features) to enhance ferry transit prediction.
- We apply the best-known ML techniques with proper processing practices for improved results.
- To gain insights into the crucial features, we employ state-of-the-art explainable artificial intelligence (AI) techniques to analyze the most important features in two different case studies.

The remainder of the paper is organized as follows: In the next section (Section 2), we present a literature review. Section 3 is dedicated to the data acquisition and construction process. In Section 4, we showcase delay importance and then illustrate the adopted methodology for delay prediction and analysis. Section 5 presents the experiments regarding ML prediction. Section 6 presents the additional analysis of the most important features affecting ferry delays. Finally, conclusions are presented in Section 7.

2. Literature review

In this section, we aim to review related literature concerning the prediction of delays, particularly in the context of arrival or travel time prediction. This encompasses studies focusing on predicting delays, as the core of these problems revolves around forecasting deviations from scheduled arrival or travel times. Notably, ferries present a unique case as they function both as vessels and transit systems, thus necessitating a tailored approach that addresses the intricacies of both domains.

We commence by elucidating how various data sources are leveraged to enhance transit systems. A recent comprehensive survey in Ge et al. (2021) provides valuable insights into this domain. Predicting transit delays entails grappling with a multifaceted and dynamic challenge, encompassing factors such as traffic demand, supply, incidents, weather, and human behavior. Effective delay prediction necessitates access to reliable and timely data from diverse sources, including GPS, sensors, cameras, and social media. Disturbances, encompassing a myriad of factors, significantly contribute to transit delays, as highlighted in Ge et al. (2022).

ML has emerged as a particularly promising approach for addressing such challenges, leveraging diverse factors – referred to as features – as inputs to yield accurate predictions. ML has demonstrated superior performance over conventional methods in numerous instances (e.g., Shanthappa et al., 2024). Notably, one of ML's key strengths lies in its capacity to assimilate heterogeneous and voluminous data sources (Jevinger et al., 2024), which is particularly advantageous in the context of delay prediction for both maritime vessels and transit systems. Indeed, ML is widely considered an appropriate approach for modeling complex datasets that exhibit non-linear relationships and missing data (Karlaftis and Vlahogianni, 2011). Consequently, we opt for ML approaches in this paper. (We note that statistical methods also still hold promise as indicated in, e.g., Tang et al., 2020).

Nevertheless, the efficacy of ML methods hinges largely on the identification and inclusion of the most relevant features. Subsequently, we outline potential data sources, taking into consideration the unique characteristics of ferries as both transit vehicles and maritime vessels.

On the one hand, many transit agencies are now publishing open data to analyze delays. Specifically, they are adopting strategies to address service disruptions by issuing updates on transit schedules using the General Transit Feed Specification (GTFS) standard and updating their policies accordingly. Over the past decade, GTFS has emerged as an industry standard for publishing data about transit operations.[4] However, providing accurately predicted arrival time information remains an ongoing challenge in the literature, and the reliability of provided predictions has been questioned on several occasions.

In addition to GTFS, other data sources have a significant impact on delay prediction, as illustrated below. In Chondrodima et al. (2022), GTFS data is utilized to construct a framework for predicting estimated times of arrival in public transport. This approach aligns with the findings of Ge et al. (2021), who emphasize that by integrating different data sources, information from one source, such as GTFS, can be corroborated by others, leading to mutually derived insights. This is exemplified in the work of Wang and Zhang (2019), who utilize mixed datasets comprising weather, train delays, and train schedules to understand patterns of train delays and predict delay times. Furthermore, Shoman et al. (2020) propose a framework that combines GTFS data with traffic data for bus delay prediction. In addition, e.g., computer vision may be considered as exemplified in Abdelhalim and Zhao (2025). In our study, we focus on weather, ridership, and vessel-related features.

On the other hand, numerous studies have explored the factors influencing ship delays. Delays represent a significant concern for both shipping operators and port authorities, making the study of delay prediction crucial for early detection and mitigation. Ship delays can entail substantial economic and environmental consequences. For a comprehensive overview of the importance of real-time delay detection and predicting vessel movement patterns until arrival, we refer to Heilig et al. (2020). Understanding the causes of delays and devising strategies to reduce them is essential. These strategies encompass traditional statistical and ML approaches, as well as advanced techniques such as deep learning and data fusion. Yan et al. (2021) offer a recent survey providing insights into various approaches for predicting ship arrival. From the survey, it becomes evident that ML methods hold significant promise in addressing these challenges. Another recent work adopting ML for vessel prediction can be found in Yu and Voß (2023).

The significance of ferry prediction in a transit system lies in optimizing schedules and minimizing waiting times, as highlighted in studies such as the survey by Gkiotsalitis et al. (2022). For instance, Liu and Miller (2020) emphasize the importance of employing appropriate optimization techniques for effective delay minimization.

Müller-Hannemann et al. (2022) utilize ML to approximate public transport schedule robustness, saving computation time while demonstrating the dependency of robustness on delay management strategies. Ricard et al. (2022) identify key features influencing transit time

[4] https://mobilitydata.org/

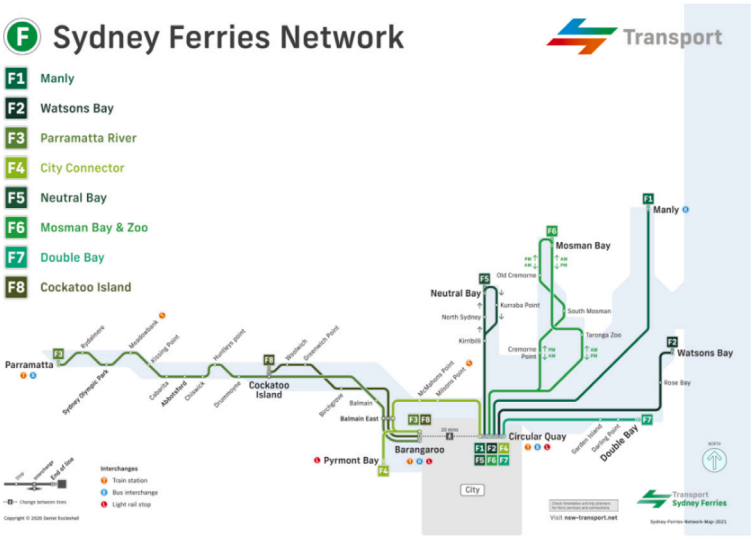


Fig. 1. Map of the Sydney ferry system.
Source: Transport for NSW.

variability and develop accurate models for predicting secondary delays, facilitating scheduling adjustments through optimization. Similarly, Parslov et al. (2023) propose two approaches for uncertainty estimation in bus travel times, suitable for transfer synchronization and connection assurance. Advanced methods integrating prediction into optimization processes are also noted by Elmachtoub and Grigas (2022), enhancing vehicle and crew scheduling as demonstrated by Ge et al. (2024).

An important question that arises is which ML algorithm to use. Different ML models, such as DNN, GB, RF, and SVR, are commonly employed to enhance delay prediction accuracy in both transit and ship operations (Müller-Hannemann et al., 2022). While linear regression has historically been utilized for delay prediction (Sun et al., 2003), recent attention has shifted to more advanced approaches like DNN, GB, RF, and SVR due to their superior performance. Smola and Schölkopf (2004) is chosen for its effectiveness in handling high-dimensional data and its ability to model non-linear relationships through the use of kernel functions. SVR is the suitable version of support vector machines when dealing with continuous variables such as delays. DNN are selected for their powerful capability to learn complex and non-linear patterns from large datasets. DNNs consist of multiple layers of interconnected neurons, allowing for hierarchical feature extraction (Zhang et al., 2019). Third, GB is utilized due to its proficiency in improving predictive accuracy by combining weak learners into a strong ensemble model. This iterative technique minimizes the residual errors of the combined learners (Mason et al., 1999). Lastly, RF is employed for its robustness and ability to handle large datasets with high dimensionality. RF operates by constructing multiple decision trees during training and outputting the mode prediction (regression) of the individual trees (Breiman, 2001).

The most suitable method often depends on the specific case study (Sarhani and Voß, 2024). For instance, RF models demonstrate high accuracy for short-term train delay prediction (Li et al., 2020). DNN outperformed other methods in a different case study (Nithishwer et al., 2022), while SVR consistently performs well across multiple studies, including the work of Müller-Hannemann et al. (2022). GB models have been shown to effectively capture train arrival delays and railway system characteristics (Shi et al., 2021), and were found to be the most effective in the preliminary work of Sarhani et al. (2023), which is directly relevant to this research.

Lastly, understanding the importance of different features is crucial. Explainable AI techniques are rapidly improving to provide more interpretable models, aiding decision-making tasks. These methods mark a milestone in ML research, enhancing our understanding of the reasons behind the results, which is crucial in our context. Examples of related work in this regard are Sariyer et al. (2024) and Li (2023).

From the literature, it is clear that there is a necessity to utilize state-of-the-art ML for delay prediction and analysis of ferry transit to enhance synchronization. In this paper, we employ these methods to address this issue, marking, to the best of our knowledge, the first comprehensive approach to do so. To better highlight research gaps and position our contributions, we classify the relevant literature in Table 1, summarizing key studies, data sources, methodologies, and findings.

3. Data

The goal of this section is to outline the various datasets utilized for predicting transit delays for both the Sydney and the New York case studies. We begin by presenting an overview of the case studies, followed by an explanation of the data collection process and the diverse features considered.

3.1. Case studies

Our first case study examines the transit ferry network serving Sydney, Australia. Fig. 1 provides a visual representation of the network. (More information can be found at Transport for NSW.) The second case study focuses on New York City (NYC), where NYC Ferry serves as a public network of ferry routes. Fig. 2 provides a visual representation of the network. (More information can be found at NYC Ferry Routes and Schedules.)

As shown in Figs. 1 and 2, the Sydney ferry network consists of eight routes, while NYC operates six routes along with an additional service. Despite these slight differences, both ferry systems exhibit comparable characteristics and operate on a similar scale. We note that in the NYC case study, the arrival delay is not directly available but can be derived by comparing the actual arrival with the scheduled time.

In this study, various data sources are used to analyze possible transit delay patterns and to identify the factors affecting these delays. We begin by describing the internal data source features provided by the agencies through their portals, followed by a description of the external open data sources collected from other sources.

Table 1
Summarized literature review (selection).

Category	Key studies	Findings
Motivation	Gkiotsalitis et al. (2022), Liu and Miller (2020)	Transit prediction is needed for synchronization and service optimization.
Data sources	Ge et al. (2021), Wang and Zhang (2019)	GTFS, ridership, weather, and historical delays are key inputs for delay prediction.
Transit prediction	Shi et al. (2021), Shoman et al. (2020)	ML frequently outperforms traditional approaches in transit delay prediction.
Vessel prediction	Heilig et al. (2020), Yan et al. (2021), Yu and Voß (2023)	ML has gained prominence in vessel delay prediction.
ML models	Müller-Hannemann et al. (2022), Ricard et al. (2022), Parslov et al. (2023)	Performance of RF, GB, SVR, DNN, and other models varies across contexts.
Feature importance	Li (2023), Sariyer et al. (2024)	Feature attribution methods are essential for practical deployment and decision making.

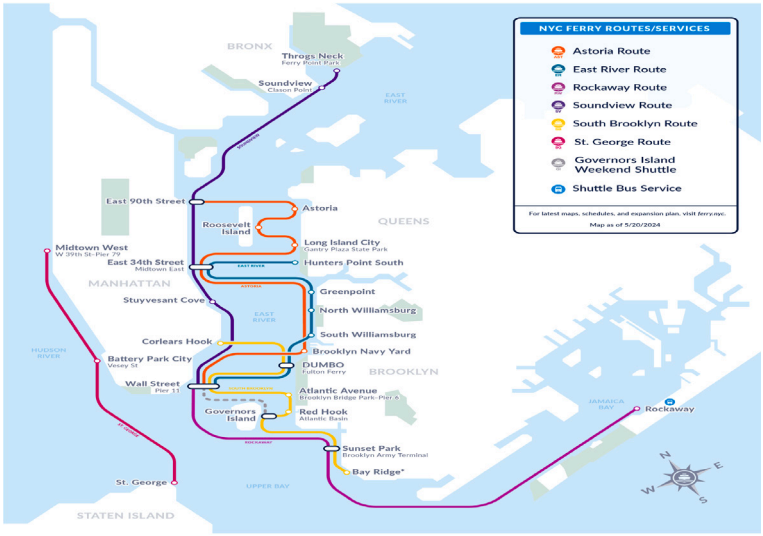


Fig. 2. Map of the New York ferry system.
Source: NYC Ferry.

3.2. General transit feed specification data

The primary internal data source is the GTFS, an open standard used to distribute relevant transit system information to riders. We apply GTFS data from both available categories, GTFS static and GTFS real-time.

We begin with the target variable, which in our case is arrival delay. As defined in the GTFS specification, these updates typically provide a predicted arrival or departure time for stops along the route. However, several agencies also offer historical data on arrival and prediction times. In the case of our studied data, arrival and departure delay information is provided after the events have occurred. This was verified by comparing the timestamps with actual arrival times.

In this study, we focus solely on arrival delays and do not consider other closely related variables, such as actual arrival times or departure times and delays. Considering the synchronization cases showcased in Section 4.1, studying arrival times is more important than departure times, though the proposed prediction and analysis could be generalized to departures. Therefore, dwell time is excluded from the analysis.

In addition to arrival delay, we note that agencies provide many features, but most of them are not expected to significantly impact delays (Sarhani and Voß, 2024). Examples of such features include "departure uncertainty" and "trip schedule relationship". These features provide little information on the delays and have shown that they are not significant. We decided to focus more on other factors that have previously shown a significant impact (Chondrodima et al., 2022 and Sarhani and Voß, 2024). Our aim in this study is to determine the importance of those factors, as shown below. Examples of potentially relevant information include "route ID", "trip ID", "stop ID", and "stop sequence". For Sydney ferries in GTFS data, a 'stop ID' is a unique code assigned to each ferry terminal (e.g., Circular Quay, Manly Wharf), while 'stop sequence' represents the sequential order of stops a ferry makes during its journey (e.g., Circular Quay as the starting point, followed by Manly Wharf as the next stop, etc.). We also include features about the vehicles, namely "vehicle ID" and "vehicle label" (this information is often part of the vehicle position feed, while the aforementioned features and the target variable are part of the trip update feed of GTFS static).

We note that dwell time, as mentioned above, is not directly incorporated. However, related factors such as weekdays are included, as dwell time depends on the number of passengers, which, in turn, is influenced by weekdays. We have added two features: one about

weekdays and one binary feature about holidays (indicating weekends or public holidays).

To extract GTFS data, we adopted the NSW Open Data Portal (the data is available at NSW Open Data Portal) (accessed on September 23, 2024), which contains both GTFS static and GTFS real-time data. For GTFS real-time, we used the free API provided by the website to extract the real-time information. Due to its ability to provide comprehensive open data, this portal is increasingly adopted for research, and the Sydney case study was recently used in the study by Young et al. (2023). For the NYC case study, we used data from the NYC ferry service, which is available at NYC Ferry Developer Tools (accessed on September 23, 2024).

3.3. Other data sources

In addition to the GTFS data, agencies in both cases provide information about ridership and vessel capacity. Moreover, weather, an additional external data source not provided by agencies, is necessary in this case.

3.3.1. Ridership and vessel capacity data

For the Sydney case study, information about ridership and vessel capacity is available on the aforementioned NSW Open Data Portal (accessed on September 23, 2024). In this case, we added ridership information provided by the portal about the number of passengers using the OPAL card each month (the portal provides the numbers for each type, e.g., adult, child, and we combined them into one feature to simplify the analysis). We also included a class feature and a vessel capacity feature, as they can impact the importance of ridership on the vessels.

For the NYC case study, the ridership information is available at NYC Ferry Ridership (accessed on September 23, 2024). This dataset reports hourly ferry boardings by route, direction, ferry landing, and day of the week. We used the day as a merging key for integration with other features. Regarding vessel capacity, the vessels are categorized into two groups based on their capacity: either 150 or 350 passengers, as indicated on that portal.

3.3.2. Weather data

In addition to extracting relevant information provided in the portals, we added several weather features. Weather is another real-time information source valuable in many fields. As indicated in the literature, weather factors that can affect transit include temperature, precipitation, and wind. Such information can influence the number of passengers and delays.

Many APIs have been proposed for weather data retrieval. In this paper, we used hourly weather data for the city of Sydney and of New York from Wunderground (accessed on September 23, 2024). Specifically, for each hour of the day on each trip observation, we added the corresponding weather information provided on that website. Considering the limited available information on the impact of weather on ferry delays, it is unclear which weather features are the most important. Therefore, we included all the features provided on the website and analyzed their importance in Section 6.

3.4. Description of the features

In this section, we summarize the features used in this study and describe them in Table 2. That is, Table 2 illustrates the different features used in the study. We start with GTFS features, followed by ridership and class features, and then weather features. We note that the features Headway, Dwell Time, Turnaround Time, and Route Length are not available directly but they can be extracted from the available information. Moreover, Dwell Time is not considered in the New York case as all recorded values were 0. For merging purposes, we added other features such as the hour of the day, but here we are focusing only on the features used for prediction and analysis.

3.5. Data pre-processing

Before working with the data, the first step is data pre-processing. An important issue when processing agency data, as pointed out in Barbeau (2018), is its accuracy. Some agencies might fill in data with random values just to fill gaps instead of providing missing values.

In this paper, we first removed missing values and then removed outliers in the prediction phase (any value outside the 3 z-score range). Another issue is eliminating various redundant columns in the merged data frame.

For Sydney, the data used in this study covered the period from 31 January 2022 to 02 January 2024. We collected over 1 million observations (1,323,620), while each observation corresponds to a specific trip, stop, or vehicle, as we removed information providing redundant details. After outlier removal, 1,237,770 observations were kept. For NYC, the data used in this study covered the period from 21 April 2024 to 05 May 2024 and from 19 September 2024 to 04 October 2024 (109,224 observations). After outlier removal, 104,025 observations were kept. We have uploaded the code and a link to the data on GitHub.[5]

4. Methodology for predicting and analyzing ferry delays in transit systems

This section emphasizes the need for delay prediction in the context of transit synchronization, highlighting its importance for optimizing schedules. We then describe the ML process, and outline the experimentation setup, including model training and tuning. Finally, we outline the process of using Shapley values for feature importance analysis.

4.1. Motivating delay prediction through transit synchronization

Before delving into the prediction of delays with ML, we showcase the need for it by analyzing Sydney data. To optimize the transit network and synchronization, deterministic approaches can be used. However, delays are quite common for transit ferries, especially during inclement weather and rush hours. Additionally, there are instances where ferries arrive before the planned time, especially if a buffer was included in the scheduled arrival time.

In the data analyzed here, we found that ferries arrived 3 min after the scheduled time in 13.28% of the cases, and 3 min before the scheduled time in 1.10% of the cases.[6] Therefore, leveraging ML predictions is crucial in optimizing the transit schedule and determining buffer times between ferries and other modes of transportation.

This section explicitly demonstrates the necessity of delay prediction and analysis to enhance public transport networks. Specifically, we address the challenge of optimizing ferry-to-bus synchronization, a critical concern for ferry users aiming to align with connecting buses efficiently. We illustrate this by comparing scenarios where planning relies solely on static schedules versus those incorporating ML predictions. Our focus is on assessing the impact of these approaches on passenger waiting times. While acknowledging the complexity of synchronizing transit comprehensively, as detailed in Ge et al. (2022), our objective here is to showcase the practical advantages of ML methods through three case studies.

In instances of ferry delays, passengers risk missing subsequent transit connections, primarily buses. While rapid transit frequencies, such as those observed on Sydney's ferry route 1 and tram 8, may mitigate this issue, lower frequency services significantly impact passenger satisfaction.

[5] More detailed information is available on GitHub (https://github.com/Malek-01/Ferry-transit-delay-).

[6] The information in this section is provided by extracting the information available at https://transportnsw.info/trip#/trip and analyzing the available data accordingly.

Table 2
Description of the features used in the study.

	Feature	Description
GTFS	Stop Sequence	The order of a stop for a particular trip
	Trip Date	The starting date of the trip
	Trip Start Time	The starting time of the trip
	Vehicle Label	A visible label for the user
	Headway	Time gap between consecutive trips on the same route
	Route Length	Length of the ferry route
	Dwell Time	Planned Time spent by the ferry at a stop before departure
	Turnaround Time	Time between a ferry's arrival at its final stop and departure for the next trip
	Direction ID	Indicates travel direction of the ferry (e.g., inbound or outbound)
	Number of Stops	Indicates the number of stops of a trip
Vessel & Ridership	Ridership	Total number of passengers boarding the ferry during a specific period, aggregated from ticketing data
	Class	Vessel class
	Total Capacity	Passenger capacity of the ferry
	Weekday	Day of the week of the trip
	Holiday	Binary feature indicating if the day is a weekend or a public holiday
	Peak Hours	Indicates whether a trip occurs during peak or off-peak hours
Weather	Wind	Wind direction recorded each hour
	Wind Speed	Wind speed recorded each hour
	Wind Gust	Wind gust recorded each hour
	Pressure	Pressure recorded each hour
	Condition	Weather condition recorded each hour
	Humidity	Humidity recorded each hour

Two main scenarios adversely affect passenger satisfaction: inadequate buffer time leading to missed connections and excessive buffer time resulting in prolonged waiting periods. In cases of insufficient buffer time, exemplified by ferry route 6 (stop 3) and bus line 111 with a 2-minute buffer, delays of at least 2 min occur in 10.5% of the instances. Given the bus's 30-minute frequency, this percentage undermines customer satisfaction. Increasing buffer times, considering that instances where the ferry arrives early are very rare (0.01%), can alleviate prolonged waiting periods. Conversely, excessive buffer times, as seen in the ferry route 8 (stop 4) and bus line 538 connection with a 10-minute buffer, result in passengers waiting longer than necessary. With 99.21% of the ferry instances experiencing delays of less than 5 min, the risk of missing connections is very small. However, in rare cases (0.22%), delays exceed 10 min; yet, this poses little threat to connection reliability. Thus, we recommend decreasing buffer times in such situations.

While acknowledging the need for a comprehensive transit schedule update to support these statements, our focus remains on emphasizing the importance of incorporating ferry delay prediction into transit planning. Notably, existing research, such as Müller-Hannemann et al. (2022), leverages ML for optimizing transit schedules, while ongoing work, as outlined in Ricard et al. (2022), contributes to this evolving field.

In the subsequent sections, we delve into the predictive and analytical aspects of delay analysis using ML, laying the groundwork for future endeavors toward improved synchronization.

4.2. Experimentation setup

To implement various algorithms, we employed the Scikit-learn library (Pedregosa et al., 2011) (version 1.0.2).

First, parameter tuning involved cross-validation, a standard technique for adjusting predictive model hyper-parameters. We conducted a grid search using GridSearchCV with 5-fold cross-validation and the RepeatedKFold method to identify the best hyper-parameter combination, minimizing the negative mean absolute error for the dataset. For example, for SVR, we used 5-fold cross-validation to optimize the values of C (regularization factor), the kernel, and ϵ on the validation set. Due to the data length for the case of Sydney, the cross-validation phase is performed on a sample of the data. Table 3 presents the hyper-parameter values considered by GridSearchCV.

The chosen hyper-parameters for different methods are presented in Table 4. Default parameter values provided by Scikit-learn were retained for other parameters. For DNN implementation, we employed Keras 2.2.5 with TensorFlow as the backend, fixing the number of epochs at 50.

Second, for the comparison of the methods, in order to have a fair assessment of the ML prediction phase, the data must be divided into a training set and a test set. In our experiment, the first 80% of the data are used for training and the remaining 20% are used as test data. Our aim in using this division is to see how accurate we can get predictions on the (unseen) test data. The training of SVR involves the use of feature selection, selecting features based on their importance scores (Butcher and Smith, 2020), as it has improved the accuracy in Sarhani and Voß (2024). The implementation of the algorithms and processing can be found in the same Github repository.

All the experiments are carried out on a computer equipped with an Intel i7-9750H and 16 GB of RAM. The measures adopted in this paper are: R^2 (coefficient of determination), the root mean square error (RMSE). Additionally, following the approaches in Nair et al. (2019) and Kim et al. (2017), we define a threshold to evaluate the percentage of cases where the difference between the predicted and actual delay values is within an acceptable range. In this study, we consider a 1-minute (60-second) threshold for acceptable predictions (predictions within 1-min). Furthermore, we compare the computational time (CPU time) required by the different methods, where the processing time covers the entire training and testing phases.

4.3. Feature importance analysis using shapley values

The aim of this section is to leverage the SHAP explainable AI technique to explore the most important features. This technique is crucial for stakeholders to understand and trust model predictions, as it provides insight into the factors most influencing ferry delays, such as weather conditions and peak-hour congestion. This method was recently adopted in similar research by Sariyer et al. (2024), Li (2023), and Wagner et al. (2022). These three papers used explainable AI techniques to understand the impact of various transportation modes, including public transportation, ridesharing, and sustainable urban mobility, and the importance of various factors.

For SHAP analysis to provide meaningful insights, it must be based on accurate model predictions. In this paper, SHAP analysis was performed on a GB model. The results achieved using solely the open data were strong, with an R^2 greater than 0.65 for Sydney and R^2 greater than 0.75 for NYC. As in Wagner et al. (2022), it is possible to draw important conclusions as the obtained results in our case are better. To

Table 3
Hyper-parameter search space for grid search.

Model	Initial grid search space
SVR	$C \in \{0.1, 1, 100\}$, $\epsilon \in \{0.1, 0.2, 0.5\}$, Kernel: {linear, rbf, poly, sigmoid}
RF	$n_estimators \in \{50, 100, 200\}$, max_depth: {None, 10, 20, 30}
GB	$n_estimators \in \{50, 100, 200\}$, learning_rate: {0.01, 0.05, 0.1}, max_depth: {3, 5, 7}
DNN	Activation: {relu, tanh}, Regularization $\alpha \in \{0.0001, 0.001, 0.01\}$, Solver: {Adam, sgd}, Learning Rate: {constant, adaptive}, Init Rate: {0.001, 0.01, 0.1}

Table 4
Chosen hyper-parameters for machine learning models with cross-validation.

Algorithm	Parameter	Sydney value	NYC value
SVR	Kernel	Linear	RBF
	Regularization Parameter (C)	100	100
	Epsilon (ϵ)	0.1	0.1
DNN	Optimizer	Stochastic Gradient Descent (Learning Rate = 0.001)	Adam (Learning Rate = 0.001)
	Activation Functions	Hyperbolic Tangent ($tanh$)	ReLU
	Loss Function	Mean Absolute Error	Mean Absolute Error
	Regularization	L2 Regularization (λ = 0.001)	L2 Regularization (λ = 0.001)
	Additional Layers	Batch Normalization, Dropout (0.5)	Batch Normalization, Dropout (0.5)
RF	Number of Estimators ($n_estimators$)	100	150
	Max Depth	5	10
GB	Number of Estimators ($n_estimators$)	50	200
	Learning Rate	0.1	0.1
	Max Depth	7	7

Table 5
Statistical summary of arrival delay for Sydney (in seconds).

Year	Count	Mean	Std Dev	Min	Q1	Median	Q3	Max
2022	352,786	26.33	60.16	−123	0.0	0.0	54.0	205
2023	699,789	27.38	63.02	−123	0.0	0.0	62.0	205

further validate the conclusions, we also performed SHAP analysis on other algorithms, and the key findings remained consistent across the models.

4.4. Delay analysis distinguishing between station and run delays

Table 5 presents the statistical summary of the *arrival delay* variable for the years 2022 and 2023. Delays are measured in seconds, with positive values indicating late arrivals and negative values indicating early arrivals. The raw data excludes outliers using the 1.5 IQR (interquartile range) rule. It should be noted that analyzing outliers is important, as they may represent actual extreme scenarios that need to be addressed (as showcased in Section 4.1 and discussed in Fig. 3).

To distinguish between station delays and run delays, we computed the dwell time delay (i.e., departure delay minus arrival delay). The mean dwell time delay is 22.90 s, while the median is 0 s. This difference indicates that the majority of the stops experience negligible dwell time delays, with a few extreme cases driving up the mean. Additionally, comparing these values with Table 5 confirms that most delays accumulate during stops.

Similarly, Table 6 presents a summary of arrival delays for the NYC case.

By computing the actual dwell time, we determined that the median dwell time is approximately 2.6 min (156 s) and the mean is about 2.8 min (168 s). This indicates that, similar to the Sydney case, a significant portion of delays accumulates during stops, underscoring the importance of accurately accounting for dwell times.

It is also noteworthy that the planned dwell time in the NYC dataset is uniformly set to zero—possibly reflecting the assumption that passengers are primarily interested in departure times. However, omitting accurate arrival times poses challenges for addressing synchronization issues, as discussed in Section 4.1.

Moreover, Fig. 3 is designed to illustrate the average arrival delays for Sydney's ferry services in 2022 and 2023 (without outlier removal), with the objective of highlighting temporal and route-specific patterns. Panel A ("Arrival Delay by Hour") focuses on how delays vary throughout the day, while Panel B ("Arrival Delay by Route") compares delay performance across different routes.

The analysis reveals a significant spike in average arrival delays between 15:00 and 18:00, indicating that the most severe delays occur during the late afternoon. This suggests a strong temporal pattern, probably related to peak travel demand.

Route-level comparisons show moderate variations in delays, with certain services exhibiting increased lateness in 2023 relative to 2022. These results suggest that factors such as peak periods and route characteristics are key contributors to overall delay accumulation. Given the available data, no consistent interpretation can be drawn; the observed delays are likely influenced by a combination of factors such as number of stops, route length, and vessel capacity. These aspects are explored in more detail for both case studies in Section 6.

5. Experimental results for the prediction of ferry delays

In this section, we present and discuss the results for delay prediction. It is important to note that, considering this is the first work on predicting ferry transit delays, there is no baseline model available for comparison.[7]

5.1. Correlation of the features and the ferry delays

5.1.1. Sydney case

We begin the experiments by analyzing the correlation between various features and delay (in seconds). The two case studies yielded almost similar results regarding the correlated features, and in Fig. 4 we present the heatmap for the Sydney case.

[7] The only work addressing this is Sarhani et al. (2023). This study builds upon it by using the same methods, incorporating a much larger dataset for the first case study to produce more reliable results, and adding a second case study to facilitate more meaningful decision making.

Table 6
Statistical summary of arrival delay (in seconds) for NYC.

Year	Count	Mean	Std Dev	Min	Q1	Median	Q3	Max
2024	84,204	188.40	2049.60	−2451.60	−641.40	−383.40	−119.40	6935.40

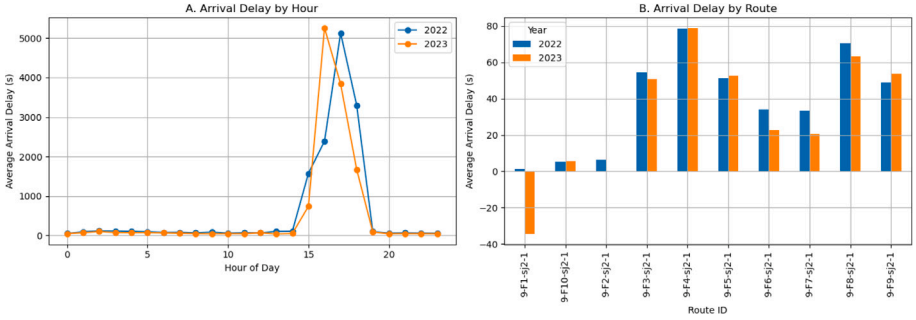


Fig. 3. Comparison of average arrival delays by hour and route for the Sydney case study.

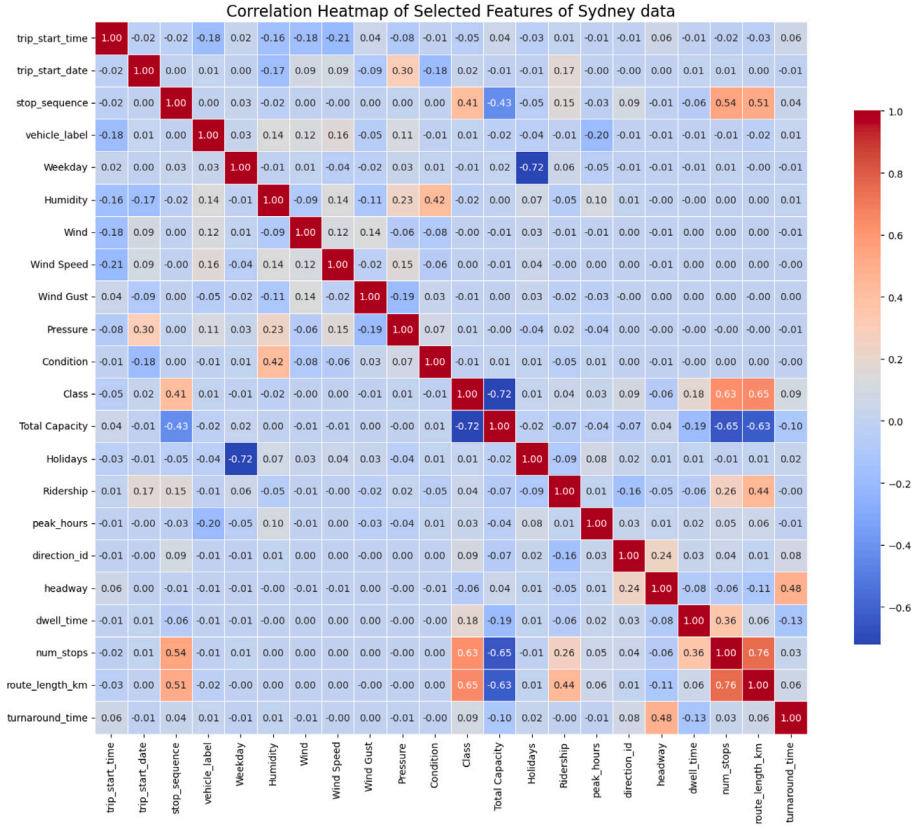


Fig. 4. Correlation plot of the features used for delay prediction for the Sydney case study.

The correlation heatmap (Fig. 4) for the Sydney dataset reveals several expected relationships among the features. Strong positive correlations are observed among operational variables such as the number of stops, route length, and turnaround time. In contrast, Total Capacity shows a negative correlation with both Class and Holidays. Weather-related variables exhibit generally low correlations with other features, suggesting a limited direct impact on operational characteristics.

Notably, Route Length is highly correlated with Number of Stops, raising concerns about potential multicollinearity. To address this, we follow the approach proposed by O'brien (2007), which recommends eliminating the features that exhibit the highest correlation with others. Applying this method, we removed Number of Stops, Pressure, and Humidity from the Sydney dataset.

These insights highlight the need for a more detailed analysis. Given the complexity and multifactorial nature of the relationships between these variables, traditional correlation analysis may not fully capture the underlying patterns and interactions. This motivates the use of ML techniques, which can model nonlinear relationships and interactions more effectively.

5.1.2. New York case

In the New York case (Fig. 5), similar patterns to the Sydney case can be observed, with most correlations being weak. Noteworthy associations include a moderate negative correlation between headway and direction ID, and between route length and number of stops, suggesting that route structure and directionality may influence operational variables. Additionally, Weekday and Holidays show a positive correlation,

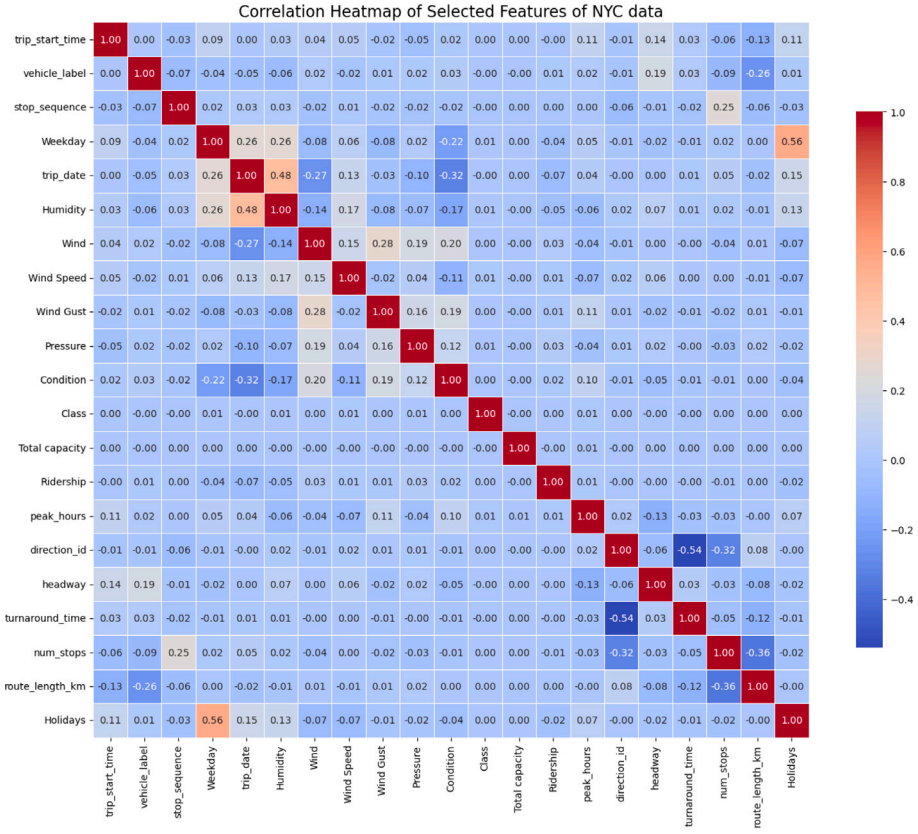


Fig. 5. Correlation plot of the features used for delay prediction for the NYC case study.

Table 7
Comparison of ML predictions for the Sydney transit delay prediction.

Metric	SVR	GB	DNN	RF
R^2	**0.7321**	0.6588	0.6366	0.5467
RMSE	**252.36**	278.41	294.77	312.38
Predictions within 1-min	**92%**	89%	84%	80%
CPU time (s)	3198.72	**380.62**	1816.44	395.18

Table 8
Comparison of ML predictions for the New York transit delay prediction.

Metric	SVR	GB	DNN	RF
R^2	0.7143	0.7807	**0.7814**	0.6289
RMSE	258.11	**197.13**	198.08	278.16
Predictions within 1-min	78.34%	88.39%	**90.03%**	72.07%
CPU time (s)	380.11	**53.16**	268.14	58.47

indicating potential overlap in calendar-based patterns. Overall, the complexity of these interrelations reinforces the value of using machine learning methods to better capture hidden dependencies and nonlinear effects in delay prediction. Following the procedure outlined in O'brien (2007), the variable Holidays was removed from the New York dataset due to its high correlation with other features.

5.2. Comparison of machine learning techniques for Sydney delay transit prediction

In this part, for the different ML algorithms, we present a comparison of their prediction capabilities in the test set for the city of Sydney in Table 7. This table displays the R^2, RMSE, sMAPE, and CPU time values obtained by comparing the algorithm predictions with the real values in the test set.

From Table 7, we observe some slight differences in the predictive abilities of the ML algorithms. Despite the inability to achieve optimal results due to the unavailability of economic and infrastructure data, which are not openly accessible, all the algorithms produced robust outcomes, demonstrating the effectiveness of the parameter tuning discussed in Section 4.2. Notably, our percentage of predictions within 1 min is higher than that reported in Nair et al. (2019). However, Nair et al. (2019) dealt with the more complex Deutsche Bahn's nationwide railway network, which presents greater operational challenges. In contrast, our study benefits from a less intricate ferry system, enabling more precise predictions, driven by granular feature engineering that captures stop-specific factors, ridership, and environmental variables. Notably, SVR exhibited the best performance, which aligns with the findings in Müller-Hannemann et al. (2022), further confirming the effectiveness of thorough parameter tuning. However, this improved accuracy comes at the cost of increased computational complexity, with SVR requiring significantly longer CPU times, particularly when working with large datasets, such as ours, which contains over a million observations. This illustrates a significant trade-off between predictive accuracy and computational efficiency that must be carefully considered when choosing an algorithm for practical applications.

5.3. Comparison of machine learning techniques for New York delay transit prediction

In the same way as in Section 5.2, we present in Table 8 a comparison study of the different methods.

We can conclude from Table 8 that DNN demonstrates the highest predictive accuracy for the NYC dataset. However, GB offers the most balanced trade-off between performance and computational cost, making it suitable for scenarios where both speed and predictive accuracy are critical.

Table 9
Performance comparison of traditional statistical models and GB for Sydney and NYC.

Metric	City	MLR	GLM	QR	KDE	GB
R^2	Sydney	0.1264	0.1621	0.0423	0.0000	0.6588
	NYC	0.1397	0.2165	0.0552	0.0000	0.7807
RMSE	Sydney	422.51	422.51	411.55	446.39	278.41
	NYC	439.28	439.28	430.12	467.84	197.13
Predictions within 1-min (%)	Sydney	70.82	70.82	14.81	8.32	89.00
	NYC	72.04	72.04	18.93	11.15	88.39
CPU time (s)	Sydney	11.47	23.15	321.22	1.94	380.62
	NYC	0.42	4.51	89.56	0.0625	53.16

In summary, given the results of both case studies, GB stands out as a favorable option for short-term transit delay prediction. It provides results comparable to the best-performing methods in both cases, with significantly lower CPU-time requirements, making it a more efficient choice for practical deployment.

5.4. Comparison with conventional statistical methods

To further contextualize the performance of machine learning approaches, we conduct a comparison with four traditional statistical methods: Multivariate Linear Regression (MLR), Generalized Linear Models (GLM), Quantile Regression (QR), and Kernel Density Estimation (KDE). These models are commonly used as baselines in delay prediction studies and provide interpretable frameworks for benchmarking machine learning performance. MLR is a standard method for modeling the linear relationship between multiple predictors and a continuous outcome, implemented using ordinary least squares (OLS). GLMs extend MLR by allowing for more flexible error distributions; in our case, we use a Gaussian family, making GLM a natural extension of MLR with the same identity link function.

QR enables the estimation of conditional quantiles of the delay distribution. In this study, we report results for the 25th percentile ($q = 0.25$), following the methodology in Tang et al. (2020), where QR was shown to outperform certain ML approaches in transportation applications. KDE is used as a non-parametric probabilistic method. Although it does not produce point predictions by design, we extract the mode of the estimated distribution to enable fair comparison across methods. The KDE implementation uses a Gaussian kernel with bandwidth set to 2.0, following Ricard et al. (2022).

All models were trained and tested on the same data split used for the machine learning models. Table 9 presents the performance metrics, including R^2, RMSE, the percentage of predictions within a 1-minute error margin, and CPU time.

The results confirm that GB substantially outperforms all traditional models across both cities. Among the statistical models, GLM performs consistently better than MLR and QR in terms of R^2, with similar or slightly improved RMSE and computational efficiency. QR and KDE show limited predictive performance due to their design focus on distributional estimation rather than precise point predictions.

6. Analysis of feature importance using Shapley values

The aim of this part is to analyze the importance of different features and to identify the most significant ones for the prediction process.

6.1. Analysis of feature importance for delay prediction in the Sydney case study

In Fig. 6, we present the bar plot of the Shapley values depicting the importance of the most significant features for the case of Sydney. Shapley values have gained widespread use in feature importance analysis, and a bar chart provides a simple yet effective way to visualize feature importance.

Fig. 6 illustrates the relative importance of various features in predicting ferry delays in Sydney. Ridership emerges as the most influential factor, underscoring the strong association between passenger demand and service punctuality. Route Length also ranks highly, reflecting the increased likelihood of delays on longer ferry routes. Vehicle Label holds significant predictive value, suggesting that vessel-specific characteristics substantially influence delay patterns. Trip Start Date and Trip Start Time highlight the temporal dynamics of delay occurrence, capturing both seasonal and daily operational variations. Headway demonstrates a notable effect, indicating that the spacing between successive trips impacts delay outcomes. In contrast, weather-related variables – including Wind Gust, Wind Speed, Wind, and Condition – exhibit relatively low SHAP values, implying that environmental factors contribute less to short-term delay variations.

Overall, the dominance of internal operational and scheduling characteristics suggests that optimizing these aspects could play a key role in reducing ferry delays in Sydney.

6.2. Analysis of feature importance for delay prediction in the New York case study

The aim of this part is to replicate the Sydney feature importance analysis in the case of New York City to confirm the findings. The SHAP values bar plot is shown in Fig. 7.

The SHAP values graph in Fig. 7 for predicting delays in the New York transit system reveals the prominence of specific features. Stop Sequence stands out as the most critical factor, underscoring how the position of a stop within a trip can significantly influence delay. Trip Start Time and Headway follow closely, reflecting the importance of both the time of day and spacing between vehicles. Vehicle Label and Turnaround Time also contribute, highlighting the impact of vessel identifiers and operational scheduling.

Ridership, Number of Stops, and Route Length capture how passenger load, trip structure, and route distance affect delays, although they are less influential compared to Stop Sequence and time-related factors. Among the weather variables, Humidity and Pressure show moderate importance, while Wind Speed, Wind, and Condition have only marginal effects. Overall, these findings indicate that operational features dominate delay predictions, making them more informative than environmental conditions in this context.

6.3. Lessons learned and recommendations

The SHAP value analyses for both the Sydney ferry and New York City transit systems reinforce the dominance of operational and temporal features in predicting delays. In both contexts, variables such as stop sequence, trip start time, headway, vehicle label, and turnaround time consistently emerged as the most influential predictors. These findings align with earlier delay analyses (Section 4.1 and Fig. 3), which revealed pronounced variations in delays across different hours and routes—particularly during peak periods.

In Sydney, ridership was identified as the most critical feature, indicating that passenger demand is a key driver of delays, as supported by the sharp afternoon delay peaks observed in Fig. 3. Similarly, in the NYC case, the prominence of stop sequence and trip start time underscores the importance of precise scheduling and effective stop management.

In contrast, weather-related variables (e.g., wind, humidity, pressure, and wind gusts) consistently showed lower predictive power, suggesting that while environmental conditions may affect transit performance under extreme circumstances, their day-to-day impact is modest compared to internal factors.

We can conclude the necessity of optimizing services, as these key predictors are internal to the agencies and can be directly improved. For example, Müller-Hannemann et al. (2022) provide an inspiring case

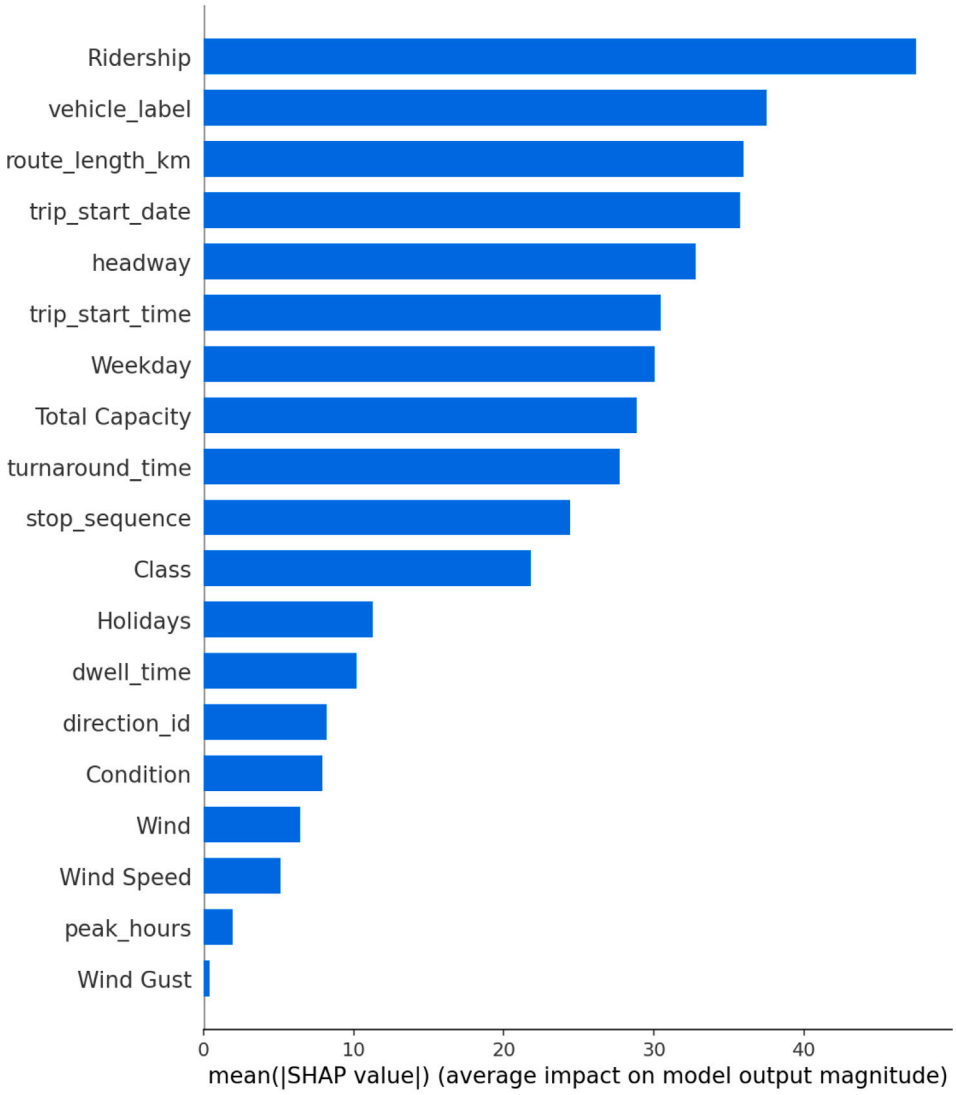


Fig. 6. Significant features impacting Sydney transit delay: SHAP value bar plot.

for other transit modes, and further insights are available in the review by Gkiotsalitis et al. (2022).

It is also important to note that, although both datasets are sourced from widely used and trusted platforms, certain anomalies raise concerns regarding data reliability. For instance, the NYC dataset sets planned dwell times uniformly, and both analyses omit the stop location type—a feature optional in the GTFS specification but potentially critical given the observations. While statistical methods were applied to handle outliers, further validation is recommended. The growing relevance of data validation is evident in emerging tools such as the MobilityData GTFS Validator[8] developed in Barbeau (2018) and subsequently enhanced.

These results lead to several key recommendations:

- **Prioritize Operational Optimization:** Enhance vehicle scheduling, turnaround planning, and stop-level operations to reduce delays. Minimizing dwell times at high-ridership stops and optimizing stop sequences can significantly improve punctuality.
- **Leverage Demand and Temporal Data:** Given the high influence of ridership and time-of-day variables, implement real-time demand forecasting and dynamic scheduling during peak hours to better align services with actual conditions.
- **Ensure Data Quality:** Transit agencies should continuously validate and refine their GTFS data. For instance, the absence of a reliable stop location type and accurate planned dwell times, as observed in the NYC case, can hinder effective delay analysis. Incorporating more precise and complete data into GTFS real-time feeds will enhance both operational management and delay predictability.

7. Conclusion

Transit delay prediction has become a crucial topic in the field of intelligent transportation systems. By predicting ferry delays using machine learning, transit authorities can better determine appropriate buffer times, thereby reducing missed connections and excessive waiting periods.

[8] https://gtfs-validator.mobilitydata.org/

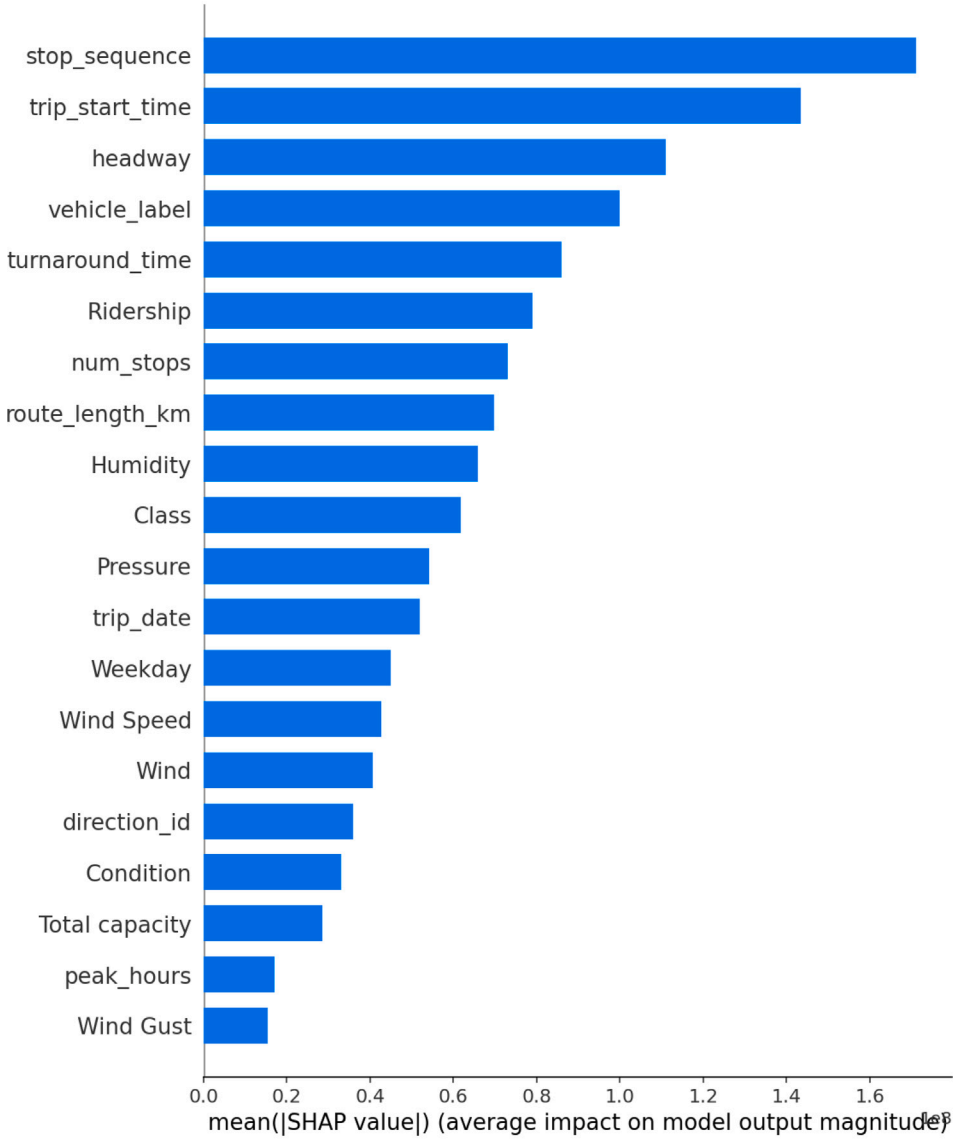


Fig. 7. Significant features impacting New York ferry transit delay: SHAP value bar plot.

Delays in transit services result from the interplay of various factors, including operational variables, environmental conditions, and passenger dynamics. This study proposes a systematic approach to predicting ferry transit delays by leveraging publicly available General Transit Feed Specification (GTFS) timetables, real-time GTFS service feed data, weather information, and ridership data. These heterogeneous sources are integrated to estimate time deviations in ferry arrivals, with machine learning techniques applied to improve prediction accuracy.

Our findings demonstrate that integrating open data sources through machine learning can significantly enhance delay prediction, even when perfect accuracy is not attainable. The methodology is adaptable and can be implemented by transit agencies worldwide, given the widespread use of GTFS standards. Moreover, by addressing known inconsistencies in GTFS feeds – such as the absence of planned dwell times in the NYC case – this study contributes to improving the practical utility of such data for prediction tasks.

Overall, while support vector regression offered the highest accuracy in Sydney, its computational cost limits its suitability for real-time applications. Gradient boosting, on the other hand, strikes a strong balance between performance and efficiency across both cities, making it a more practical option for real-world deployment. These findings support the development of more responsive and accurate transit delay prediction systems.

The SHAP value analyses and performance metrics for both Sydney and New York confirm that operational and temporal features – such as stop sequence, trip start time, headway, and vehicle label – consistently play the most influential role in predicting delays. In contrast, weather-related variables showed limited impact across both cases, reinforcing that internal scheduling and operational characteristics are primary drivers of delay. This underscores the necessity of optimizing services, as these key features are internal to the agencies and can be directly improved. For instance, Müller-Hannemann et al. (2022) provide a compelling example for other transit modes, and additional insights are available in Gkiotsalitis et al. (2022).

Future research should compare these approaches with alternative advanced machine learning and statistical methods, such as transfer learning and Bayesian hierarchical modeling. Additionally, exploring the underlying causes of delays in greater depth – such as socio-economic or service planning factors, as suggested by Bree et al. (2020) – could offer a more holistic perspective. Further analysis of ridership patterns and their interaction with delay dynamics may also support more effective demand management and scheduling strategies. Moreover, incorporating other machine learning approaches, such

Table 10
Descriptive statistics for numerical features in Sydney and NYC.

Feature (Unit)	City	Min	Mean	Std	Max
Turnaround Time (min)	Sydney	10.00	27.13	6.41	28.00
	NYC	0.00	1.52	3.21	29.00
Headway (min)	Sydney	0.00	0.00	0.00	0.00
	NYC	0.00	19.21	13.73	45.00
Ridership per Route (monthly)	Sydney	2755.00	97 671.51	53 610.43	110 895.00
Ridership per Route (hourly)	NYC	0.00	23.08	42.95	165.00
Route Length (km)	Sydney	2.81	9.98	6.53	24.56
	NYC	5.68	14.47	7.68	28.15
Humidity (%)	Sydney	8.00	64.00	19.25	97.00
	NYC	22.00	61.56	17.55	98.00
Wind Speed (mph)	Sydney	0.00	13.67	6.38	40.00
	NYC	0.00	10.71	4.22	33.00
Wind Gust (mph)	Sydney	0.00	1.79	7.91	36.00
	NYC	0.00	2.84	7.85	40.00
Pressure (inHg)	Sydney	29.14	30.01	0.22	30.58
	NYC	29.80	30.02	0.13	30.17
Trip Start Time (hh:mm:ss)	Sydney	00:00:00	13:34:47	04:37:16	23:50:00
	NYC	00:00:00	12:27:33	08:24:09	23:59:59

as reinforcement learning, which aim to both predict and optimize outcomes, could provide additional insights.

CRediT authorship contribution statement

Malek Sarhani: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Abtin Nourmohammadzadeh:** Writing – review & editing, Validation, Resources, Methodology, Formal analysis, Conceptualization. **Stefan Voß:** Writing – review & editing, Validation, Supervision, Resources, Conceptualization. **Mohammed EL Amrani:** Software, Data curation.

Consent to participate

Not applicable.

Ethics approval

Not applicable.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. Feature distributions

To complement the main analysis and support reproducibility, we provide detailed statistical distributions of all features used in model development. This includes both numerical and categorical variables for the Sydney and New York City case studies. These distributions inform our understanding of the data and contribute to feature interpretation and selection.

A.1. Numerical feature distributions

Table 10 reports the descriptive statistics for all numerical features used in the study. For each feature and city, we present the minimum, mean, standard deviation, and maximum values. These statistics help to understand the variability, scale, and typical range of each feature across the two transit systems.

Table 11
Top two categorical values and their frequencies for Sydney and NYC datasets.

Feature	Value (s)	Sydney (%)	NYC (%)
Direction	0 (outbound) / 1 (inbound)	58.75 / 41.25	51.65 / 48.35
Holidays	No / Yes	71.73 / 28.27	80.38 / 19.62
Peak Hours	No / Yes	64.56 / 35.44	92.80 / 7.20
Dwell Time	0 / 60 s	30.89 / 69.11	100 / 0
Weekday	Tuesday / Wednesday	14.76 / 14.74	–
	Thursday / Saturday	–	18.46 / 17.38
Class	First Fleet / RiverCat	28.73 / 22.02	–
	WS / DB	–	14.18 / 14.18
Condition	Fair / Mostly Cloudy	39.61 / 17.05	– / 31.77
	Cloudy / –	–/–	24.40 / –
Number of Stops	2/3	29.89 / 16.88	–
	1 / 2	–	55.73 / 16.58
Stop Sequence	9 / 6	36.04 / 25.68	–
	7 / 3	–	35.39 /26.63
Total Capacity	393 / 400	28.73 / 25.14	–
	350 / 150	–	50.15 / 49.85
Wind Direction	S / W	12.41 / 9.32	–
	NE / ENE	–	22.93 / 16.34
Trip Start Date	01/01/2023/02/01/2023	0.50 / 0.36	–
	26/09/2024/28/09/2024	–	7.20 / 5.48

A.2. Non-numerical feature distributions

Table 11 presents the top two most frequent values for each discrete non-numerical feature (e.g., categorical variables) in both cities, except for "Vehicle Label", which was excluded as it consists of arbitrary identifiers with no meaningful interpretation. This summary provides insight into typical service patterns, operational settings, and weather conditions. Some values differ across cities due to local configurations (e.g., vessel types or wind directions). For features such as *Weekday* and *Condition*, additional categories exist in the data, but only the two most frequent values are reported here for brevity.

Data availability

The data and the code are available on GitHub.

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